

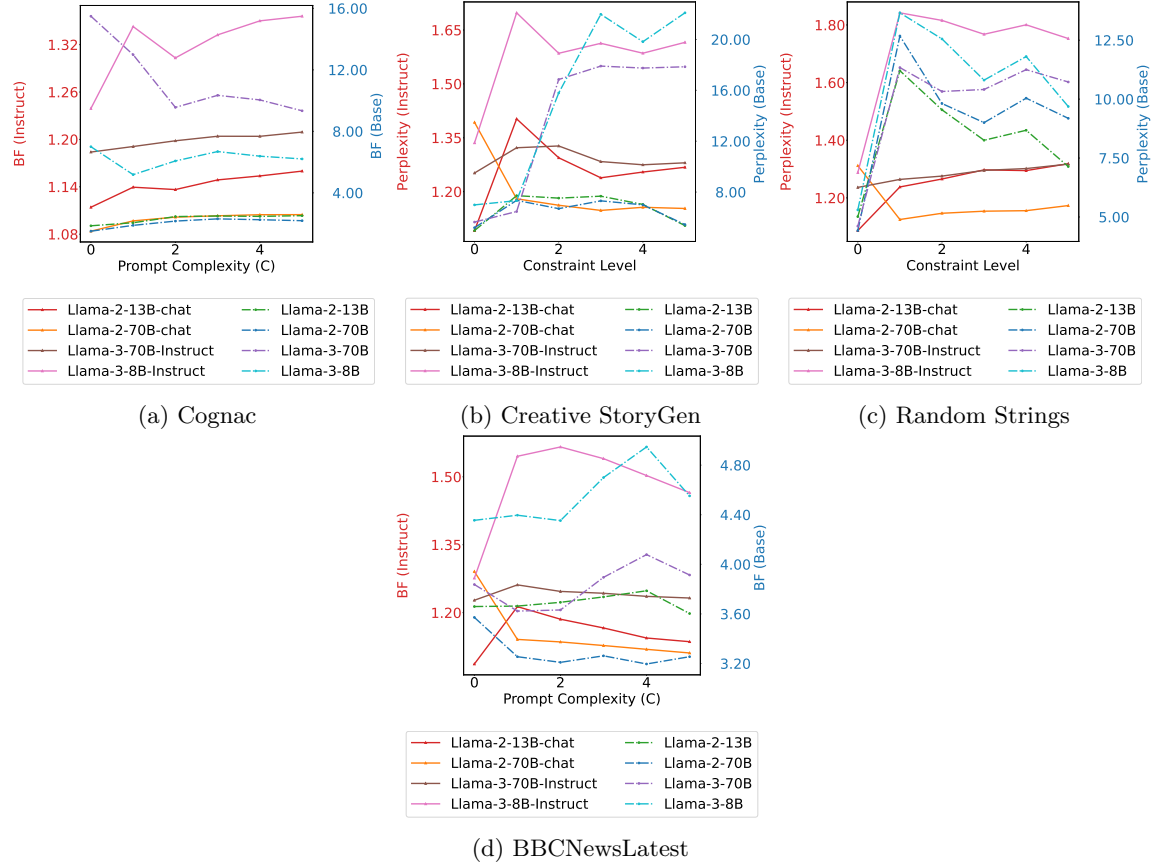


Figure 12: BF changes with prompt complexity (C) for Different Tasks. We can see prompt complexity affects BF in a task-varied way.

The key observation here is that under current computation architecture, the probability implemented by transformers are log-precision (Merrill & Sabharwal, 2023), and thus $|\log P(y_{1:N}|x;\theta)|$ is bounded (e.g., $|\log P(y_{1:N}|x;\theta)| \leq M$). For the truncated probability $\tilde{P}(y_{1:N}|x;\theta)$, we can essentially only consider the non-zero probability over the truncated vocabulary, and the same bound holds. Depending on the quantization scheme implemented, examples of M include 32, 64, etc.

We define the length-averaged *realized entropy* for a specific sequence $y_{1:N}$ as:

$$h_{\text{realized}}(y_{1:N}) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{t=1}^N H(Y_t|[x, y_{<t}]; \theta) \quad (10)$$

where $H(Y_t|[x, y_{<t}]; \theta) = -\sum_{y \in V} \tilde{P}(y|[x, y_{<t}]; \theta) \log \tilde{P}(y|[x, y_{<t}]; \theta)$.

We aim to bound the probability that the NLL deviates from this realized entropy. Let Δ_N be the total difference between the log-probability and the realized entropy sum:

$$\Delta_N = (-\log \tilde{P}(y_{1:N}|x;\theta)) - \sum_{t=1}^N H(Y_t|[x, y_{<t}]; \theta) = \sum_{t=1}^N Z_t \quad (11)$$

where we define the single-step deviation variable Z_t as:

$$Z_t \stackrel{\text{def}}{=} -\log \tilde{P}(y_t|[x, y_{<t}]; \theta) - H(Y_t|[x, y_{<t}]; \theta) \quad (12)$$

Note that Δ_N is a random variable formed by the sum of Z_t . To use Chebyshev's inequality, we need to calculate the variance of this sum:

$$\text{Var}(\Delta_N) = \text{Var}\left(\sum_{t=1}^N Z_t\right) = \sum_{t=1}^N \text{Var}(Z_t) + \sum_{i \neq j} \text{Cov}(Z_i, Z_j) \quad (13)$$

We now show that the covariance terms $\text{Cov}(Z_i, Z_j)$ are zero for all $i < j$. By definition, $\text{Cov}(Z_i, Z_j) = \mathbb{E}[Z_i Z_j] - \mathbb{E}[Z_i] \mathbb{E}[Z_j]$.

First, observe that the expected value of the deviation Z_j at any step, conditioned on the prompt and generated history, is zero:

$$\begin{aligned} \mathbb{E}[Z_j|[x, y_{<j}]; \theta] &= \mathbb{E}_{y_j} [-\log \tilde{P}(y_j|[x, y_{<j}]; \theta)] - H(Y_j|[x, y_{<j}]; \theta) \\ &= H(Y_j|[x, y_{<j}]; \theta) - H(Y_j|[x, y_{<j}]; \theta) = 0 \end{aligned} \quad (14)$$

This implies $\mathbb{E}[Z_j] = 0$ for all j . Thus, $\text{Cov}(Z_i, Z_j) = \mathbb{E}[Z_i Z_j]$.

For $i < j$, the value of Z_i is fully determined by the history $[x, y_{<j}]$. We use the Law of Iterated Expectations, conditioning on the history up to step j :

$$\mathbb{E}[Z_i Z_j] = \mathbb{E}_{[x, y_{<j}]} [\mathbb{E}[Z_i Z_j|[x, y_{<j}]; \theta]] \quad (15)$$

$$= \mathbb{E}_{[x, y_{<j}]} [Z_i \cdot \mathbb{E}[Z_j|[x, y_{<j}]; \theta]] \quad (\text{since } Z_i \text{ is determined given } y_{<j}) \quad (16)$$

$$= \mathbb{E}_{[x, y_{<j}]} [Z_i \cdot 0] \quad (\text{by Eq. 14}) \quad (17)$$

$$= 0 \quad (18)$$

Since all cross-terms are zero, the variance of the sum is simply the sum of the variances:

$$\text{Var}(\Delta_N) = \sum_{t=1}^N \text{Var}(Z_t) \quad (19)$$

Given the observation that transformer probabilities are computed with bounded log-precision (Merrill & Sabharwal, 2023), we have $|\log P(y|[x, y_{<t}]; \theta)| \leq M$ (For un-truncated P). Consequently, the random variable Z_t is bounded, and its variance is bounded by a constant, denoted $C = (2M)^2$.

$$\text{Var}(\Delta_N) \leq N \cdot C \quad (20)$$

We can now apply Chebyshev’s inequality to the length-averaged deviation:

$$P\left(\left|\frac{\Delta_N}{N}\right| \geq \epsilon\right) \leq \frac{\text{Var}(\Delta_N/N)}{\epsilon^2} = \frac{\frac{1}{N^2}\text{Var}(\Delta_N)}{\epsilon^2} \leq \frac{\frac{1}{N^2}(N \cdot C)}{\epsilon^2} = \frac{C}{N\epsilon^2} \quad (21)$$

Taking the limit as $N \rightarrow \infty$, the probability of deviation approaches 0. Thus, we have convergence in probability:

$$-\frac{1}{N} \log \tilde{P}(y_{1:N}|x; \theta) - h_{\text{realized}}(y_{1:N}) \xrightarrow{P} 0 \quad (22)$$

I Full Nudging Experiment Results

Due to space limits, we put the nudging experiment results for MMLU here. Though on MMLU, nudging does not reduce BF that quickly as over Just-Eval-Instruct, it does bring down BF of base models significantly, which verifies our hypothesis in § 6.

J BF and Information Density

Our BF measure can also be interpreted as capturing the information density that LLMs target to facilitate efficient communication (Genzel & Charniak, 2002; Jaeger & Levy, 2006; Levy, 2008; Mahowald et al., 2013; Meister et al., 2021; Verma et al., 2023). Prior work has leveraged both token-level log-probabilities and entropy rates ($\bar{H}$) as proxies for information density in human and machine communication. In Theorem 3.1, we formalize the connection between these views, showing that BF—defined as the exponentiated entropy rate—aligns naturally with this theoretical framework. Unlike prior studies focused primarily on linguistic theory or cognitive science, our work operationalizes this principle at scale across modern LLMs, linking information density to alignment training, decoding dynamics, and output variability in a unified analysis.

K Discussion: Diversity and BF Correlation

Following the branching factor (BF) analysis in § 2, a higher BF suggests greater lexical diversity in finite samples. To examine the relationship between BF and traditional diversity metrics, we compute Distinct-N (Li et al., 2016), incorporating necessary LLM-specific adaptations (Tevet & Berant, 2021; Guo et al., 2024; Kirk et al., 2024). We then conduct a correlation analysis between Distinct-N and BF.

Our results, presented in Figure 18, show **no consistent correlation** between BF and Distinct-N. Depending on the model and task, the relationship can be strongly positive, strongly negative, or entirely absent (e.g., Llama-3-70B-Instruct on Cognac at Figure 18b). This empirical inconsistency highlights a fundamental conceptual point: *BF measures a property of the underlying probability distribution, whereas diversity metrics measure a surface property of finite samples.*

BF, as the exponentiated entropy, characterizes the “width” of the model’s entire output distribution. In contrast, metrics like Distinct-N describe a small set of sampled outputs and are known to be unreliable proxies for distributional properties, being sensitive to confounding factors like generation length (Liu et al., 2022).¹³ This distinction is critical, as two models can produce samples of similar diversity while having fundamentally different underlying distributions (e.g., with infinite KL-divergence), a nuance that BF captures but sample-based metrics miss. Therefore, our work focuses on probability concentration, measured by BF, as a more fundamental and insightful tool for understanding a model’s generative process.

Viewing alignment through the lens of BF reduction provides a unified framework that explains several disparate observations: it clarifies how alignment shrinks the generative horizon, why aligned models are less sensitive to decoding methods, and how techniques like Chain-of-Thought stabilize generation by shifting information to low-BF regions. This focus on distributional properties aligns with emerging research highlighting the importance of a model’s entropy in understanding and improving advanced reasoning capabilities Cui et al. (2025); Wu et al. (2025b).

¹³While the EAD metric (Liu et al., 2022) mitigates this issue, it remains influenced by vocabulary size and is not model-agnostic.